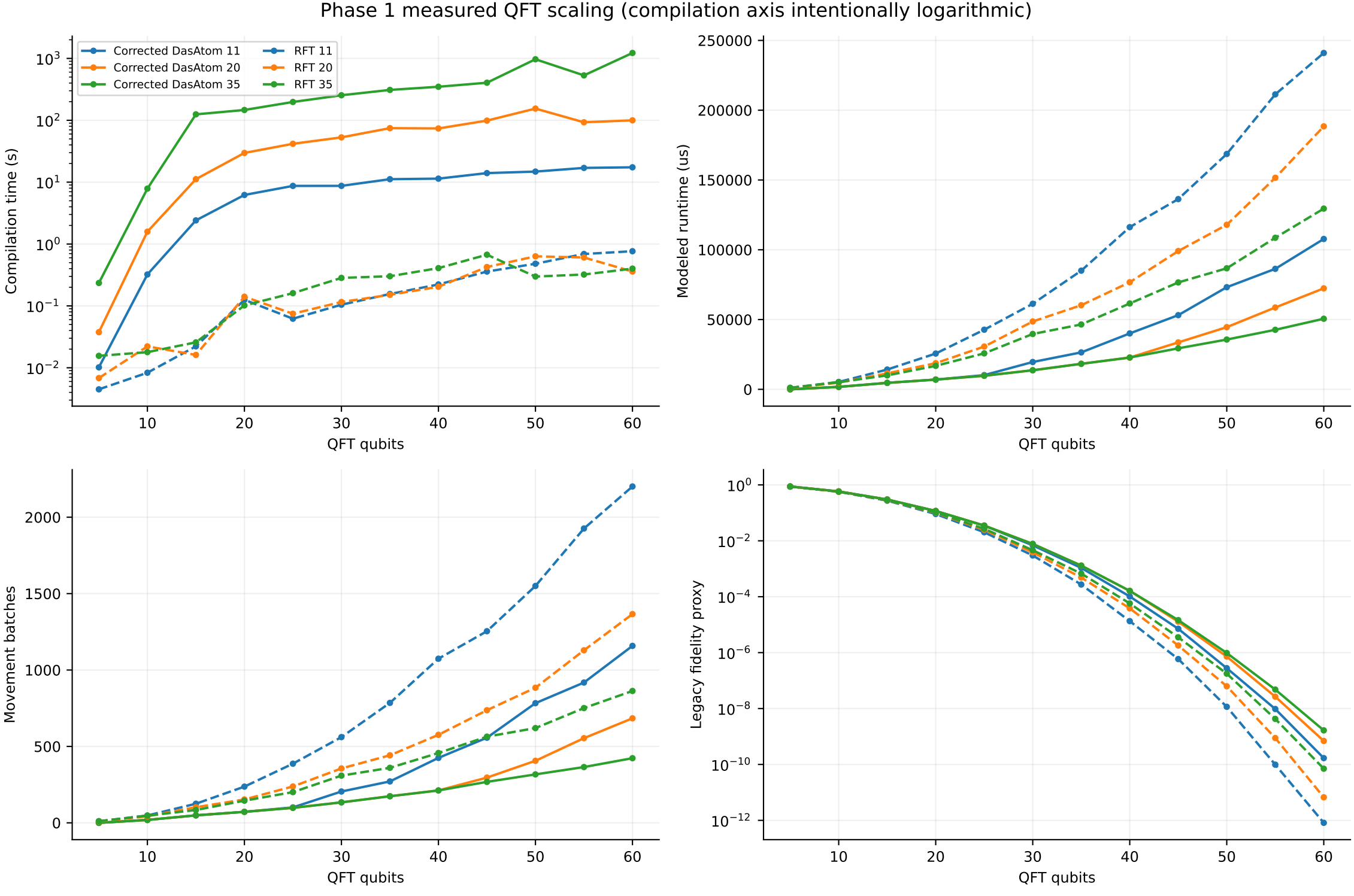

Phase 1 measured QFT scaling (compilation axis intentionally logarithmic)
Corrected DasAtom 11
Corrected DasAtom 20
Corrected DasAtom 35
RFT 11
RFT 20
RFT 35
Compilation time (s)
Modeled runtime (us)
Movement batches
Legacy fidelity proxy
QFT qubits

Paper Figure 4 vector reconstruction versus pinned-source measurements

11 x 11

20 x 20

35 x 35

Compilation time (s)

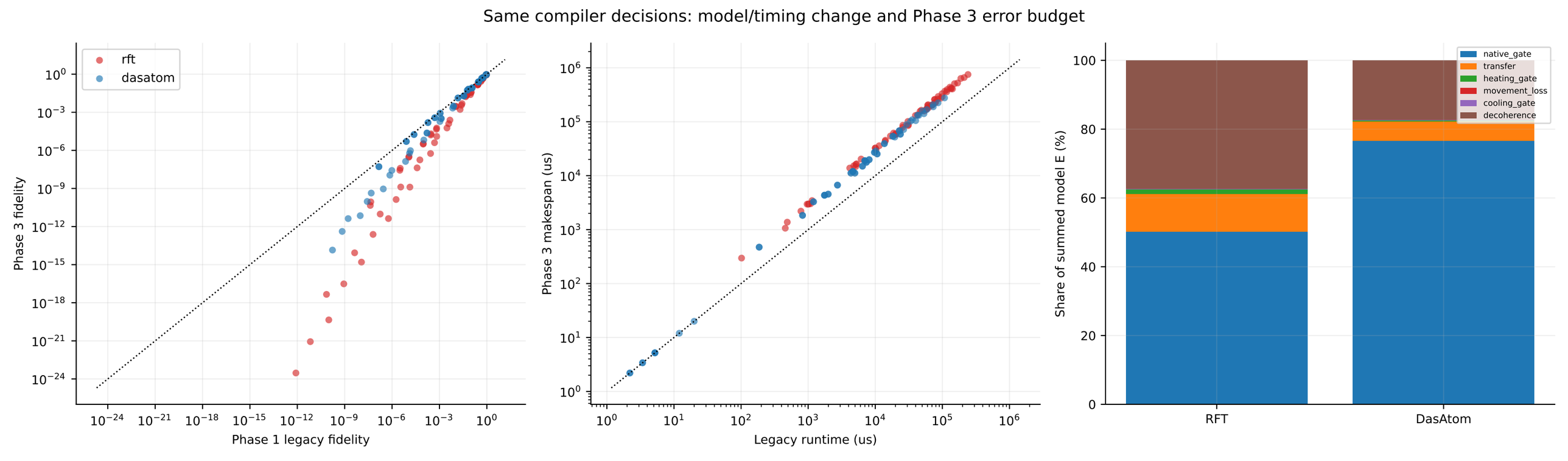
dasatom measured
dasatom paper
rft measured
rft paper

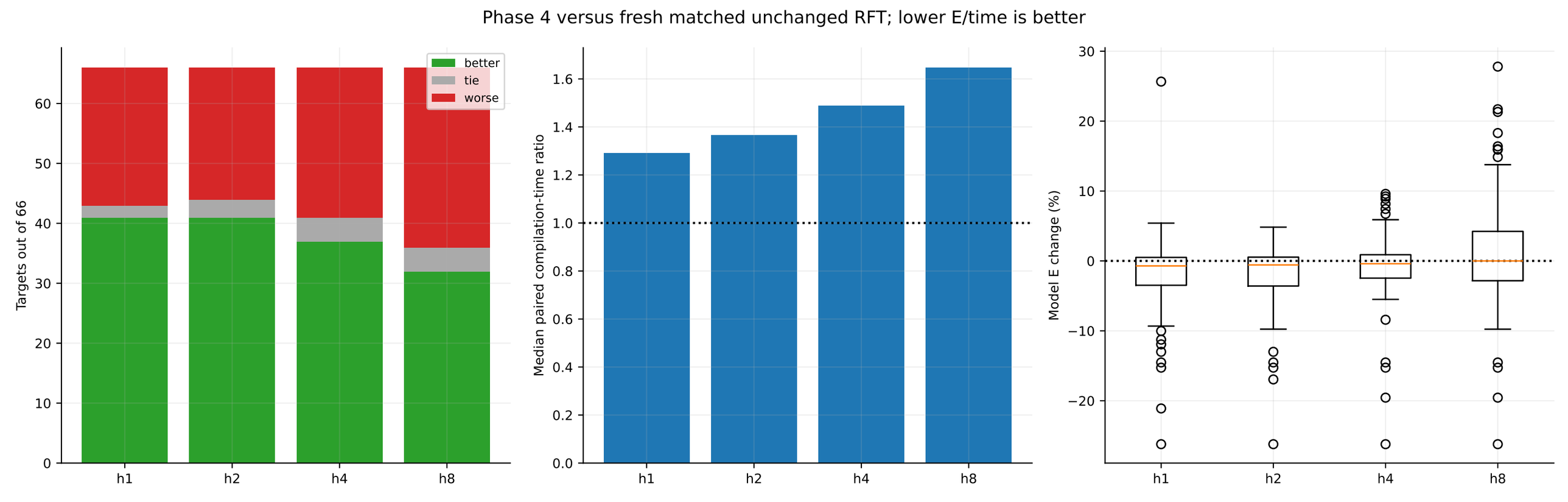
QFT qubits

Modeled runtime (us)

Movement batches

Legacy fidelity proxy


Same compiler decisions: model/timing change and Phase 3 error budget
rft
dasatom
Phase 3 fidelity
Phase 1 legacy fidelity
Phase 3 makespan (us)
Legacy runtime (us)
Share of summed model E (%)
native_gate
transfer
heating_gate
movement_loss
cooling_gate
decoherence
RFT
DasAtom


Phase 4 versus fresh matched unchanged RFT; lower E/time is better
Targets out of 66
better
tie
worse
h1
h2
h4
h8
Median paired compilation-time ratio
Model E change (%)

Regenerated Table II: pinned RFT median times and corrected DasAtom

circuit	compile s 11 dasatom	compile s 11 rft	compile s 35 dasatom	compile s 35 rft	fidelity 11 dasatom	fidelity 11 rft	fidelity 35 dasatom	fidelity 35 rft	moves 11 dasatom	moves 11 rft	moves 35 dasatom	moves 35 rft
4mod5-v1_22	0.011667	0.0028584	0.23017	0.0049653	0.94633	0.94533	0.94633	0.94177	0	1	0	5
adr4_197	2.1053	0.28917	36.511	0.26773	0.00050687	0.00028201	0.00050687	0.00029283	78	611	78	568
alu-v0_27	0.014124	0.0037532	0.23626	0.0060182	0.91829	0.90852	0.91829	0.90719	0	10	0	11
bv_n16	0.017979	0.003202	1.2452	0.0090897	0.92573	0.92276	0.92573	0.91989	2	5	2	8
cm152a_212	0.28572	0.066033	3.3327	0.077875	0.06888	0.056174	0.06888	0.056139	8	193	8	200
dc1_220	0.92045	0.1112	9.6358	0.13696	0.014928	0.011025	0.014928	0.011051	28	313	28	317
ising_model_16	0.017207	0.015332	0.24791	0.016112	0.47139	0.42251	0.47143	0.44155	0	99	0	49
misex1_241	12.003	0.20877	300.21	0.68731	2.4605e-05	1.1753e-05	2.4605e-05	1.1466e-05	84	765	84	787
qft_16	4.1671	0.014883	98.088	0.041972	0.28671	0.25853	0.28671	0.26981	46	127	46	87
qv_n16_d5	43.424	0.0046566	1430.3	0.012323	0.54102	0.51892	0.54102	0.52419	12	50	12	33
rd53_251	0.26856	0.036296	3.0031	0.14366	0.057948	0.046114	0.057948	0.047657	21	229	21	202
rd73_252	2.289	0.2384	34.239	0.7493	7.9883e-06	3.0975e-06	7.9883e-06	3.2854e-06	113	970	113	927
square_root_7	106.24	0.27658	2202.6	0.98204	1.4651e-07	4.1771e-08	1.4651e-07	4.5106e-08	252	1396	252	1131
sym6_145	0.77193	0.17159	8.5848	0.59377	0.00018801	9.3368e-05	0.00018801	9.4639e-05	52	685	52	674
z4_268	1.4156	0.086102	25.159	0.38659	0.0011133	0.00064319	0.0011133	0.00066924	69	556	69	528